

Blueprint: native-decide instances
(flt-vandiver-primes)

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August 7, 2026

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Abstract

`flt-vandiver-primes` instantiates the engine of `flt-vandiver` with finite `native_decide` certificates, yielding `FermatLastTheoremFor` p for every prime $17 \leq p < 1000$ and for both Wolstenholme primes 16843 and 2124679. The engine assembly theorems are external leaves.

0.1 Certificate instances

Definition 0.1.1 (Engine assembly (external)). *From `flt-vandiver`: `fermatLastTheoremFor_of_certs_95` (Case II via the small witness, Case I via the B_{p-3} certificate) and its Germain variant.*

Definition 0.1.2 (Fast per-prime assembly). *`fltPrimeWitnessFast`: a one-line template turning the `native_decide` Q_i and irregularity certificates into `FermatLastTheoremFor` p at a non-Wolstenholme prime.*

Definition 0.1.3 (Germain per-prime assembly). *`fltPrimeWitnessSGSub`: the Wolstenholme variant, taking the Q_i certificate and a Sophie Germain subgroup certificate.*

Theorem 0.1.4 (First irregular prime). *`FermatLastTheoremFor` 37. All 162 primes $17 \leq p < 1000$ follow this template.*

Theorem 0.1.5 (Wolstenholme prime 16843). *`FermatLastTheoremFor` 16843 (4 Q_i slices + one SG certificate).*

Theorem 0.1.6 (Wolstenholme prime 2124679). *`FermatLastTheoremFor` 2124679 (64 Q_i slices + one SG certificate).*